F78AB Actuarial and Financial Mathematics B - Solutions to May 2014 Exam

1. (a)

$$\begin{aligned}
A(10) =& 1\,000 \exp\left\{\int_0^{10} \delta(s)ds\right\} \\
=& 1\,000 \exp\left\{\int_0^{4} \delta(s)ds + \int_4^{10} \delta(s)ds\right\} \\
=& 1\,000 \exp\left\{\int_0^{4} (0.07 - 0.005s)ds + \int_4^{10} 0.055ds\right\} \\
=& 1\,000 \exp\left\{\left[0.07s - 0.0025s^2\right]_{s=0}^{s=4} + \left[0.055s\right]_{s=4}^{s=10}\right\} \\
=& 1\,000 \exp\left\{0.24 + 0.33\right\} \\
=& 1\,768.27.
\end{aligned}$$

   (b) With $\rho(t) := 100e^{0.065t}$ paid from $t = 5$ to $t = 15$ only, the present value at time 0 is

$$\begin{aligned}
PV(0) =& \int_5^{15} \rho(t) \exp\left\{-\int_0^{t} \delta(s)ds\right\} dt \\
=& \int_5^{15} \rho(t) \exp\left\{-\int_0^{4} \delta(s)ds - \int_4^{t} \delta(s)ds\right\} dt \\
=& \int_5^{15} \rho(t) \exp\left\{-\int_0^{4} (0.07 - 0.005s)ds - \int_4^{t} 0.055ds\right\} dt \\
=& \int_5^{15} \rho(t) \exp\left\{-0.24 - 0.055(t-4)\right\} dt \\
=& \int_5^{15} 100 \exp\left\{0.065t - 0.24 - 0.055t + 0.22\right\} dt \\
=& 100e^{-0.02} \int_5^{15} e^{0.01t} dt \\
=& 100e^{-0.02} \left[\frac{e^{0.01t}}{0.01}\right]_5^{15} \\
=& 1\,083.74.
\end{aligned}$$

2. The no-arbitrage or fair forward price is

$$K^{\text{fair}} = 10e^{0.03\times 6/12} - 0.50e^{0.03\times(6/12-2/12)} = 10.15113 - 0.51010 = 9.64611.$$

Therefore, "buy low, sell high" means that we sell the replicating portfolio of the forward contract and buy the forward at forward price $K := 9.50$.

The arbitrage strategy is:

At time 0,

- short-sell 1 unit of stock from a broker for £10,
- deposit £10 from the bank,
- long in forward at forward price $K = 9.50$.

Value of portfolio at time 0 is $-10 + 10 + 0 = 0$.

At time 2/12, need to pay 50p to broker as short-sold (borrowed) 1 unit of stock

- borrow 50p from the bank. Total amount of deposit in bank is $10e^{0.03\times 2/12} - 0.50$.

At time 6/12,

- Total amount of deposit in bank is $10e^{0.03\times 6/12} - 0.50e^{0.03\times 4/12} = 9.64611$.
- Through the forward contract, buy 1 unit of stock for 9.50.
- Give 1 unit of stock to broker.

Value of portfolio at time 6/12 is $+9.64611 - 9.50 = 0.14611$.

This is an arbitrage strategy which results in a risk-free profit: the portfolio had value zero at time 0 and strictly positive value at time 6/12 with no risk of loss.

3. (a) We want to find y_2 with

$$(1+y_2)^2 = \frac{(1+y_5)^5}{(1+f_{2,3})^3} \quad \Rightarrow \quad y_2 = \left(\frac{(1.0475)^5}{(1.0425)^3}\right)^{1/2} - 1 = 0.055045.$$

(b) The issue price of the bond per £100 nominal is

$$5P_1 + 5P_2 + 105P_3 = 5(0.95) + 5(0.90) + 105(0.85) = 98.50,$$

under the assumption that there is no arbitrage in the market.

4. Denote the annual yield on the investment in year n by i_n. Then

$$i_n = \begin{cases} 0.05 & \text{with probability } 0.2 \\ 0.07 & \text{with probability } 0.6 \\ 0.09 & \text{with probability } 0.2. \end{cases}$$

As $(i_n)_n$ are i.i.d. r.v.s, we have for $n = 1, 2, \ldots, 15$,

$$E[1 + i_n] = E[1 + i_1] = 0.2(1.05) + 0.6(1.07) + 0.2(1.09) = 1.07$$

and

$$E[(1+i_n)^2] = E[(1+i_1)^2] = 0.2(1.05)^2 + 0.6(1.07)^2 + 0.2(1.09)^2 = 1.14506.$$

(a) The mean accumulation of £1 000 at the end of 15 years is

$$\mathbb{E}\left(1\,000 \prod_{n=1}^{15}(1 + i_n)\right) = 1\,000 \prod_{n=1}^{15} \mathbb{E}\,(1 + i_n) = 1\,000(1.07)^{15} = 2\,759.03.$$

(b) As

$$\mathbb{E}\left(1\,000^2 \prod_{n=1}^{15}(1 + i_n)^2\right) = 1\,000^2 \prod_{n=1}^{15} \mathbb{E}\,(1 + i_n)^2 = 1\,000^2(1.14506)^{15},$$

the standard deviation of the accumulation of £1 000 is

$$\left(1\,000^2(1.14506)^{15} - (2\,759.03)^2\right)^{0.5} = 126.42$$

at the end of 15 years.

(c) (i) As the expected interest rate is unchanged at 7% p.a, the mean accumulation is the same as in part (a).
The spread of the yields around the mean 7% p.a is higher than in part (b). Hence, the standard deviation of the accumulation will be higher than £126.42.

(ii) Mean accumulation will be more than £2 759.03 since the investment is being accumulated over a longer period.
The standard deviation of the accumulation will be higher than £126.42 since investing over a longer time period than in part (b) will lead to a wider range of possible accumulated amounts.

5. (a) As RPI(Jan 14) = RPI(Dec 13)$\times 1.03^{1/12}$ = $253.4 \times 1.03^{1/12}$ = 254.024953,

$$\frac{2}{2} \times \frac{\text{RPI(Jan 14)}}{\text{RPI(Jul 13)}} = \frac{254.024953}{249.7} = 1.0173.$$

(b) As RPI(Jul 23) = RPI(Dec 13)$\times 1.03^{9\frac{7}{12}} = 253.4 \times 1.03^{9\frac{7}{12}} = 336.379882$

$$100 \times \frac{\text{RPI(Jul 23)}}{\text{RPI(Jul 13)}} = 100 \times \frac{336.379882}{249.7} = 134.7136.$$

(c) Let $i = 0.065$ and $j := \left(\frac{1.065}{1.03}\right)^{0.5} - 1$. The indexed coupon payments are

Payment date	Apr 14	Oct 14	Apr 15	$\cdots$	Oct 23
Indexed coupon	1.0173	$1.0173(1.03)^{0.5}$	$1.0173(1.03)^{1}$	$\cdots$	$1.0173(1.03)^{9.5}$

The price of the bond at 18 December 2013 is

$$\begin{aligned}
&(1+i)^{2/12}\Big(1.0173v^{0.5} + 1.0173(1.03)^{0.5}v + 1.0173(1.03)^{1}v^{1.5} + \cdots + 1.0173(1.03)^{9.5}v^{10} \\
&\qquad + 134.7136v^{10}\Big) \\
=&(1+i)^{2/12}\left(\frac{1.0173}{1.03^{0.5}}\left[(1.03v)^{0.5} + (1.03v)^{1} + (1.03v)^{1.5} + \cdots + (1.03v)^{10}\right]\right. \\
&\qquad \left. + 134.7136v^{10}\right) \\
=&(1+i)^{2/12}\left(\frac{1.0173}{1.03^{0.5}}a^{j}_{\overline{20}|} + 134.7136v^{10}\right) \\
=&(1+i)^{2/12}\left(\frac{1.0173}{1.03^{0.5}}(16.859848) + 134.7136v^{10}\right) \\
=&89.6009.
\end{aligned}$$

(d) The money yield on the bond increases. This is because the investor gets larger money payments for the price £89.6009.

6. (a) Denote the present value of the variable annuity by $\mathcal{V}_{\overline{n}|}$. Then

$$\begin{aligned} \mathcal{V}_{\overline{n}|} &= 1^2v^1 + 2^2v^2 + 3^2v^3 + \cdots + n^2v^n \\ \Rightarrow \quad v\mathcal{V}_{\overline{n}|} &= 1^2v^2 + 2^2v^3 + 3^2v^4 + \cdots + (n-1)^2v^n + n^2v^{n+1} \\ \Rightarrow \quad (1-v)\mathcal{V}_{\overline{n}|} &= 1^2v^1 + (2^2-1^2)v^2 + (3^2-2^2)v^3 + \cdots + (n^2-(n-1)^2)v^n \\ &\quad - n^2v^{n+1}. \end{aligned}$$

As $k^2 - (k-1)^2 = 2k - 1$,

$$\begin{aligned} (1-v)\mathcal{V}_{\overline{n}|} &= (2\cdot 1 - 1)v^1 + (2\cdot 2 - 1)v^2 + (2\cdot 3 - 1)v^3 + \cdots + (2\cdot n - 1)v^n \\ &\quad - n^2v^{n+1} \\ &= 2\left(v^1 + 2v^2 + 3v^3 + \cdots nv^n\right) - v^1 - v^2 - v^3 - \cdots - v^n \\ &\quad - n^2v^{n+1} \\ &= 2(Ia)_{\overline{n}|} - a_{\overline{n}|} - n^2v^{n+1}. \end{aligned}$$

Hence

$$\mathcal{V}_{\overline{n}|} = \frac{2(Ia)_{\overline{n}|} - a_{\overline{n}|} - n^2v^{n+1}}{1-v}.$$

(b) (i)

$$\mathrm{PV}_L = 190\,000 a_{\overline{20}|} + 10\,000(Ia)_{\overline{20}|} = 190\,000\frac{1-v^{20}}{i} + 10\,000\frac{\ddot{a}_{\overline{20}|} - 20v^{20}}{i}.$$

Calculating at $\delta = 5\%$ p.a., we find $a_{\overline{20}|} = 12.32898$, $\ddot{a}_{\overline{20}|} = 12.96111$ and $(Ia)_{\overline{20}|} = 109.29192$. Hence $\mathrm{PV}_L = 3\,435\,425$.
To calculate the duration, the numerator is

$$\begin{aligned} &1 \times 200v + 2 \times 210v^2 + 3 \times 220v^3 + \cdots + 20 \times 390v^{20} \\ =&1 \times (190+10)v + 2 \times (190+20)v^2 + \cdots + 20 \times (190+200)v^{20} \\ =&190(v + 2v^2 + 3v^3 + \cdots + 20v^{20}) + 10(v + 2^2v^2 + 3^2v^3 + \cdots 20^2v^{20}) \\ =&190(Ia)_{\overline{20}|} + 10\mathcal{V}_{\overline{20}|}. \end{aligned}$$

From part (a),

$$\begin{aligned} \mathcal{V}_{\overline{20}|} =& \frac{2(Ia)_{\overline{20}|} - a_{\overline{20}|} - 20^2v^{21}}{1-v} = \frac{2(109.29192) - 12.32898 - 20^2v^{21}}{1-v} \\ =&1359.01124. \end{aligned}$$

Hence the numerator of the duration is 34 355 577 and dividing by the present value $\mathrm{PV}_L = 3\,435\,425$, gives the duration 10.000 years.

(ii) To purchase £1 nominal of the 25–year ZCB at time 0 costs $v^{25} = 0.28650$. Therefore the cost of nominal amount £X of the ZCB is $0.28650X$ today.
To purchase £1 nominal of the coupon-bearing bond today costs

$$0.06a_{\overline{12}|} + v^{12} = 0.06(8.80005) + 0.54881 = 1.07681.$$

Therefore the cost of nominal amount £Y of the coupon-bearing bond is $1.07681Y$ today.
The present value of the assets is

$$0.28650X + 1.07681Y.$$

(iii) The numerator of the duration of the assets is

$$Y\left(0.06(Ia)_{\overline{12}|} + 12v^{12}\right) + 25Xv^{25}.$$

As $(Ia)_{\overline{12}|} = \frac{\ddot{a}_{\overline{12}|} - 12v^{12}}{i}$, and $\ddot{a}_{\overline{12}|} = 9.25124$ then $(Ia)_{\overline{12}|} = 51.98836$. Hence the numerator of the duration of the assets is

$$Y\left(0.06(51.98836) + 12(0.54881)\right) + 25(0.28650)X = 7.16250X + 9.70502Y.$$

and the duration is obtained by dividing the numerator by the present value of the assets:

$$\frac{7.16250X + 9.70502Y}{0.28650X + 1.07681Y}.$$

(iv) We match present values of the assets and liabilities:

$$0.28650X + 1.07681Y = 3\,435\,425$$

and match the durations:

$$\frac{7.16250X + 9.70502Y}{3\,435\,425} = 10.000.$$

Solving, we find $X = 740\,472$ and $Y = 2\,993\,360$.